

Honors Geometry Quiz

Name: _____

Version: 1

Show all work and clearly indicate your thinking and process. Keep your work organized. Use exact values unless otherwise indicated.

1. Coordinate Quadrilateral

Graph the points:

$A(-1, 2)$

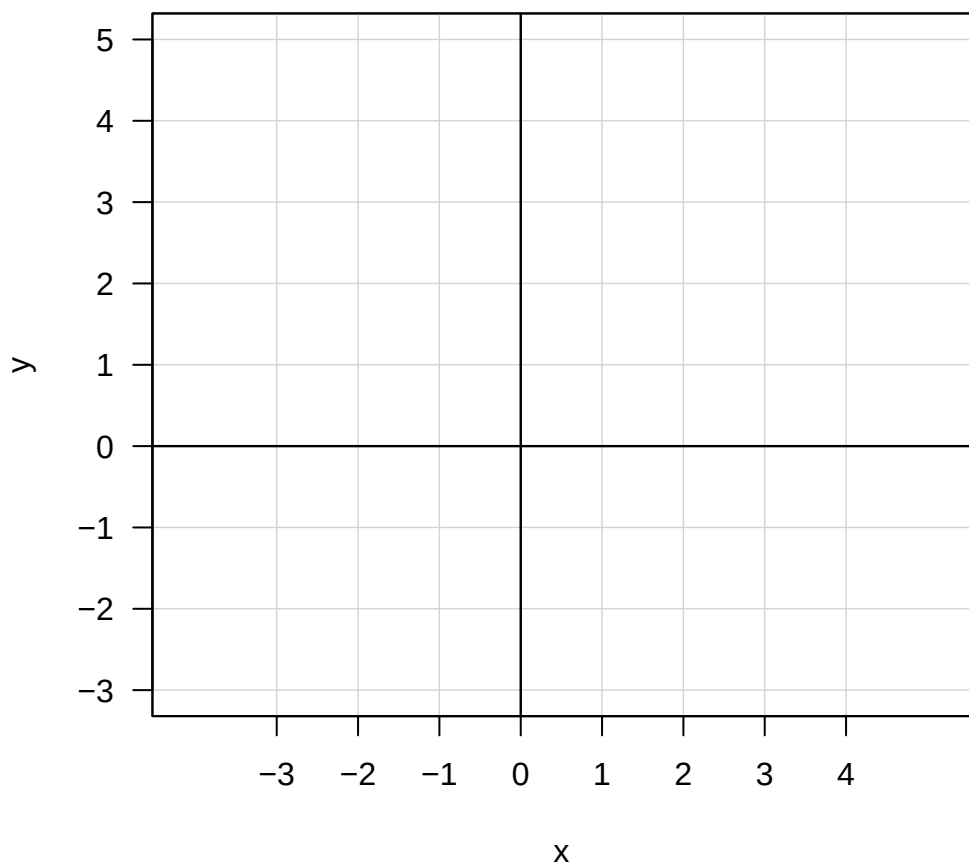
$B(1, 3)$

$C(2, 1)$

$D(0, 0)$

Is quadrilateral $ABCD$ **equilateral**?

Justify your work and explain clearly.



2. Right Triangles

Find the missing side of each **right triangle**.

The third side listed is always the hypotenuse.

Give each answer in **simplified radical form** when necessary. Show your work.

a.

5, 7, _____

b.

8, 9, _____

c.

4, _____, 7

d.

_____, 15, 17

3. Points on a Line

Find **both points** on the line

$$x = 0$$

that are

$$\sqrt{53}$$

units from the point

$$(2, 1).$$

Draw a diagram and show your work.

First point: _____

Second point: _____

4. Distance Two Ways

Given the points

$$J(-5, 1)$$

and

$$K(-8, -3),$$

answer both questions below.

a. Pythagorean Theorem

Find the length of segment JK using the **Pythagorean Theorem**.

b. Distance Formula

Find the length of segment JK using the **Distance Formula**.

5. Intersection of Two Lines

Given

$$-4x + 2y = 12$$

and

$$x + y = 3,$$

solve for their intersection.

Give your answer in **exact values** and show your work.

Intersection: _____

6. Equation of a Line

Given the points

$$(-5, 4)$$

and

$$(0, 6),$$

find the equation of the line.

Show enough work to justify your answer.

Equation: _____

7. Slope with Variables

Given the points

$$(2m + 1, 3m - 1)$$

and

$$(14m + 9, 12m + 5),$$

find the slope of the line.

Simplify completely and show your algebra.

Slope: _____

8. Algebra

a. Squaring a Binomial

Does

$$(x + 5)^2 = x^2 + 25$$

for all values of x ?

Explain.

b. Factor

Factor completely:

$$y^2 + 8y + 15$$

c. Factor Completely

Factor completely:

$$5v^2 + 35v + 30$$

9. Important Question

What is your favorite class other than Geometry?

Why?